TWENTY MINUTES IN JIANGHU: A LONG-HORIZON BENCHMARK ON AN UNMODIFIED 1996 CHINESE CRPG

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ABSTRACT

Frontier models saturate static benchmarks yet remain conspicuously bad at *playing*: perceiving an unfamiliar world, deciding what is worth doing, and making steady progress over hundreds of dependent decisions. We introduce JY-CRPG-BENCH, a benchmark that drops a vision-language agent into 金庸群俠傳 (*Heroes of Jin Yong*, Heluo Studio 1996), an unmodified commercial open-world CRPG running under DOS emulation, and gives it twenty minutes of wall-clock time. The game is a deliberately adversarial combination of capabilities that current benchmarks test only in isolation: raw-pixel perception of a low-resolution isometric world, grounding in Traditional Chinese rendered as 1996 bitmap fonts, an unconstrained keyboard action space, self-directed goal-setting with no quest log or score, and persistent character progression (levels, skills, items, reputation) that only rewards coherent multi-step play. The interface is a single curlable brief and a minimal HTTP API, so any model with any harness can play within minutes; every run is recorded, replayable, and published to a live public leaderboard. Rather than treating pixel change alone as progress, we combine request logs and validated transition detectors with inventory and character statistics read from the emulated machine's state. This distinction matters: two intuitive screen-derived progress metrics we first deployed were later proven invalid (one inflated $2\times$ by sprite orientation and menu overlays, letting a random-key baseline top the leaderboard; one structurally incapable of ever firing). Across six frontier models and a random baseline, *no agent leaves the opening scene*: every run ends at level 1 with the game's progression systems untouched, while behavioural metrics (meaningful-step ratio, oscillation, screen reads per action, think time) cleanly separate model behaviour from randomness and from each other. JY-CRPG-BENCH is not a task to be solved this year; it is an instrument for watching general agency arrive.

1 INTRODUCTION

A modern frontier model can prove competition mathematics, write working compilers, and answer expert questions in most academic fields. Placed in front of a 1996 video game that a teenager could play, given the controls, the objective, and twenty minutes, it does not manage to leave the first room. This paper is about measuring that gap carefully.

The gap is not new. BALROG found frontier models at 1.5% progression on NetHack (Paglieri et al., 2025); VideoGameBench found the best vision-language models completing 0.48% of a suite of 1990s action games (Zhang et al., 2025); ARC-AGI-3 reports frontier systems below 1% on interactive environments every tested human solves (ARC Prize Foundation, 2026). But each of these instruments removes something essential to isolate something else: BALROG's environments are research gridworlds with text renderings; TextQuests removes vision entirely (Phan et al., 2025); ARC-AGI-3 deliberately excludes language and cultural priors; VideoGameBench's games are dominated by reflexes and linear level structure, so its scoring reduces to checkpoint distance along a known walkthrough. What none of them measures is the capability stack of an *open-world role-playing game*: perceive an unfamiliar scene, decide *what is worth doing*, navigate to it, operate

Figure 1: The opening scene of 金庸群俠傳 as every agent sees it: a 320 × 200 raw frame. The protagonist stands in an isometric compound; the exit is a specific one-tile doorway; a chest (left) is the first thing worth searching. There is no quest log, no objective marker, no score. All labels, dialogue, and menus are Traditional Chinese bitmap text.

menus, read and act on in-fiction text, and accumulate persistent progress, all at once, over a long horizon, with nobody keeping score inside the game.

JY-CRPG-BENCH measures exactly that stack, using a single deep environment in the tradition of Crafter and NetHack (Hafner, 2022; Küttler et al., 2020) rather than a wide suite: 金庸群俠傳 (*Heroes of Jin Yong*), the 1996 Heluo Studio classic, run *unmodified* under DOS emulation. The game is an open world built on Jin Yong's wuxia novels: a two-tier map (one large overworld stringing together many local scenes), hundreds of named characters with the game's own statistics for level, experience, health, skills, items, and reputation, and no imposed storyline. A human first playthrough is measured in tens of hours. We give agents twenty minutes of wall clock and measure what they do with it.

Our contribution is the benchmark itself: its design, its measurement methodology, and the evidence that it is hard by nature. It is not a method for solving it:

- **A benchmark that is hard by construction, not by obscurity (§2).** The difficulty decomposes into named, independently checkable properties: raw-pixel perception of low-resolution isometric art; cross-lingual grounding in Traditional Chinese bitmap fonts; an isometric control scheme in which arrow-key intuitions are systematically wrong; a full-keyboard action space of which only a small learned subset does anything; self-directed goal-setting with no objective signal (ARC Prize Foundation, 2026); and persistent state that only rewards coherent long-horizon play.
- **A harness-free protocol (§3).** The entire agent-facing surface is one sentence, *"Read `https://hanxiao.io/jy-crpg-bench/agents.md` and play it."*, plus a two-endpoint HTTP API (send keys, read the screen). There is no SDK, no Docker image, no agent framework to adopt; any model in any harness can be playing within minutes, which sidesteps the harness-confound that plagues game evaluations (Hu et al., 2025; Phan et al., 2025). Runs are wall-clock-budgeted, isolated per session, recorded end-to-end, and published with interactive replays to a public leaderboard.
- **Ground truth from the machine, not the screen (§3.3).** Milestones and character statistics are read from the emulated machine's memory (the game's own 182-byte character records, located and validated against the game's shipped save files) and scene transitions from the renderer's fade-to-black, each detector validated with positive and negative controls. Agents never see any of this; they get pixels.
- **Two documented metric failures as a methodological warning (§4).** Our first, intuitive screen-derived exploration metric, counting distinct screens, inflated 2× under sprite ori-

entation and menu overlays, and *ranked a random-key baseline above every frontier model*. A second detector (dialogue boxes) was structurally incapable of firing: measured over 582 frames of real play, the screen region it tested never reaches the threshold, so it returned an identical zero for every run while looking like a finding. Both survived deployment and were caught only by adversarial probing; both are instances of the *outcome-validity* failures catalogued by Zhu et al. (2025), arising here not from harness bugs but from the measurement itself.

- **Baselines showing the floor is unbroken (§5).** Six frontier models (Claude Opus 5, Claude Sonnet 5, Claude Fable 5, Gemini 3.7-Flash, Grok-4.6, Qwen3.8-27B) and a seeded random-key baseline all fail to leave the opening scene. Every run ends at level 1 with zero experience: the game's entire progression apparatus goes untouched. Behavioural metrics nevertheless separate the agents sharply: meaningful-step ratio spans 0.07–0.85 against the random floor of 0.40, and screen-reads-per-action distinguishes a model that looks before every act (and still achieves nothing) from one that acts blind.

2 The game, and why it is hard by design

2.1 金庸群俠傳 in brief

Heroes of Jin Yong (Heluo Studio, 1996) is widely regarded as a milestone of Chinese-language RPGs. The player wakes as a nameless novice in a small compound, and the fiction's only standing instruction is implicit: explore the world of Jin Yong's fourteen novels, join its factions, learn its martial arts, and recruit its heroes. Concretely, the world is two-tier (many local scenes such as buildings, sects, and caves hang off one large outdoor map), and the game state carries, for each of 320 named characters, a record of level, experience, health, stamina, learned skills, carried items, reputation, and aptitude. Progress is nonlinear: there is no quest log, no waypoint, no score, and most buildings will not even open until specific items are held. The opening compound, a single scene, contains a searchable chest, furniture, walls, and exactly one exit tile.

We chose this game deliberately over better-known Western titles. It is text-heavy in Traditional Chinese, which makes language grounding a first-class obstacle rather than an accident of localisation; it is menu-driven rather than reflex-driven, so a turn-paced agent is not structurally disadvantaged (cf. Zhang et al., 2025); its progression state is rich enough to measure; and it is culturally prominent enough that walkthroughs almost certainly appear in pretraining corpora, which makes the observed failure a *knowing-doing gap* (Paglieri et al., 2025) rather than a knowledge gap, and makes the benchmark a test of acting on knowledge rather than of possessing it.

2.2 Six obstacles, by construction

Perception without concessions. Agents receive the raw 320×200 framebuffer (Figure 1), upscaled only losslessly. Sprites are $\sim$25–50 px, the palette is 256 colours, and text is 16 px bitmap CJK: min-size typography with none of the anti-aliasing modern OCR expects. Nothing is annotated, no set-of-marks, no accessibility tree (cf. Xie et al., 2024).

Cross-lingual grounding. Everything the game says (menus, dialogue, item names, place names) is Traditional Chinese. An agent that reasons in English must ground its plan ("search the chest, then find the compass") in glyphs it must read off a 1996 screen. To keep the playing field level, the brief is served in both English and Simplified Chinese, with the game's on-screen Traditional terms kept verbatim in both.

An action space with no map to the screen. The API accepts the full DOS keyboard ($\sim$90 named keys). The game responds mainly to numpad diagonals: movement is isometric, so `kp7/kp9/kp1/kp3` move along the diamond's axes and *the arrow keys' apparent directions are wrong*. The agent must discover the effective action space inside a much larger nominal one: the inverse of ARC-AGI-3's deliberately minimal five-action design (ARC Prize Foundation, 2026), and closer to how humans actually meet a machine.

Goal-setting with no signal. The game never scores, prompts, or nudges. Deciding that the chest is worth searching, that the doorway matters, that the compass gates progress: that is the agent's problem. This is the "acquire goals on the fly" demand that ARC Prize Foundation (2026) identify as central to agency, here embedded in a fiction rather than an abstraction.

Long horizon under a wall clock. Twenty minutes admits roughly 300–1,400 actions at observed cadences. The game itself is tens of hours deep; the budget makes the benchmark a measure of *early* competence: orientation, first discoveries, first progression. The wall clock, rather than a turn budget, prices deliberation: thinking for 20 s per action is a choice that costs a third of the run, an efficiency framing shared with ARC Prize Foundation (2026) and Phan et al. (2025).

Persistent state, irreversible time. The emulator runs at the game's authentic 70 Hz; there is no pause, no savescumming, no reset within a run. What the agent does is what happened.

3 BENCHMARK DESIGN

3.1 ONE SENTENCE OF ONBOARDING

The complete integration cost for a participant is pasting one line into their agent:

```
Read https://hanxiao.io/jy-crpg-bench/agents.md and play it.
```

The brief behind that URL is the whole contract: how to open a session (`POST /session` with a model name), the two calls that constitute play, the rules of a run, and orientation advice equivalent to a game manual's first page. The API surface during play is deliberately minimal:

Call	Body	Meaning
`POST /api/key`	`{"key":"kp3"}`	press one key (optional hold, repeat)
`POST /api/keys`	`{"keys":[...]}`	press several in order
`GET /api/screen`	–	current frame as PNG
`GET /status`	–	clock, counters, run state

Because the protocol is plain HTTP, "submitting to the benchmark" and "pointing your agent at a URL" are the same act. We report models by the name the agent self-declares; the equivalent of a leaderboard submission is a finished run. This design makes the benchmark harness-agnostic in the sense that Hu et al. (2025) advocate measuring: we fix nothing about memory, prompting, or reasoning strategy, and measure the model+harness system that actually shows up. The trade-off (self-reported identity, uncontrolled scaffolds) is the same one accepted by open arenas (Chiang et al., 2024), and the recording of every run keeps claims auditable.

3.2 THE PROTOCOL OF A RUN

A run is created with a playtime (default 20 minutes; the clock starts at the first *playable* moment, not at connection). Each session is a fresh, isolated emulator process with a private copy of the game: no state, no filesystem, and no save slots are shared between sessions, and 24 sessions run concurrently on one 8-vCPU host (measured headroom: 32 sessions at full 70.09 fps before memory, not CPU, becomes binding). An agent that goes ten minutes without pressing a key is torn down and recorded as idle: on this benchmark, thinking for ten minutes about one action *is* a failure mode, not a thinking style. When the budget lapses, the session renders its recording to video, computes final metrics, and appends itself to the public catalogue; the whole lifecycle is node-local with no central orchestrator. Every run gets a stable URL with a scrubbable replay (an interactive timeline of every keypress over the video: a 7–13 KB sidecar against a 118 MB raw recording), and the leaderboard is a static page reading published JSON, so spectator load never touches game hosts.

3.3 MEASUREMENT: READ THE MACHINE, NOT THE PIXELS

The agent sees only pixels. The *benchmark* reads the machine.

Character ground truth. The game's save format is a plain six-section archive whose second section is 320 records of 182 bytes, one per named character, with fields for level, experience, HP, skills, four role-local item slots, reputation, and aptitude. NPCs use those slots for item grants and non-player battle inventory; they are not player-configurable bags. The same block stores the party's 200-slot public inventory immediately before the character array; the inventory-gain milestone compares that bag with the opening state rather than confusing it with the protagonist's four role-local slots. We validated the layout against the game's own shipped save files (across all 320 records, only two violate $\mathrm{hp} \le \mathrm{maxHp} \vee \mathrm{mp} \le \mathrm{maxMp}$, both data-entry quirks of the original game) and then located the same array in the live emulated machine by anchoring on an NPC name that occurs exactly once and confirming two neighbouring records at the correct 182-byte stride. Reads cost $1.2\,\mathrm{ms}$ via serialization into a reused buffer; a cached anchor check costs $0.3\,\mu\mathrm{s}$. Crucially, measurement is *read-only*: the one write-path we tried (reloading a savestate to calibrate coordinates) crashed DOS outright inside live sessions (§4).

Scene transitions. The game changes scene through a fade to black. We sample mean framebuffer luma every frame ($1.1\,\mu$s per call) while waiting for the screen to settle after each action. Calibration: the opening scene reads luma 92, a real transition reads 0, and the threshold of 12 sits far from anything the game otherwise draws. Controls: 74 actions of walking and menu-toggling produced zero false positives; a savestate-backtracking flood-fill of the opening scene located its exit tile and confirmed the detector fires (luma 0) exactly at the transition.

Behavioural metrics. Following the game-agent literature, we track: *meaningful step ratio* (share of actions that changed the screen at all; Nasir et al., 2025), *oscillation* (A→B→A reversals; Nasir et al., 2025), *screen reads per action* (does the model look before acting), *think-time* percentiles (inter-action gaps), *time to first action*, and *action-space coverage* (distinct keys used). Each is computable from the run's own traffic, so they hold for any harness.

The milestone summary. Progress is summarised as seven observable milestones, in the spirit of Crafter's achievements (Hafner, 2022): *acted*, *screen responded*, *picked something up*, *left the opening scene*, *reached the world map*, *gained experience*, and *reached level 2*. These mix request-log, framebuffer, and machine-state evidence and are not claimed to form one strict total order: the game permits different paths. Their count is an achievement summary rather than a hand-weighted score. A milestone a run predates (instrumentation landed mid-campaign) is reported as *unmeasured*, never as failed: an unmeasured zero and a true zero are different claims, and conflating them is how one of our own metrics went wrong.

Leaderboard statistics. The meaningful-step ratio is a binomial proportion, so the leaderboard shows 95% Wilson intervals (Wilson, 1927) and *rank ranges* rather than bare ranks: models whose intervals overlap are shown as statistically unseparated, the convention modern arenas have converged on (Chiang et al., 2024). Because ratio alone rewards doing very little (one careful action scores 1.0) and count alone rewards mashing, the board also shows the quality-throughput Pareto frontier (Figure 3): a model is only shown as dominating another when it is better on both.

4 PITFALLS: TWO METRICS THAT LIED TO US

We report two measurement failures in full because both are natural designs, both were deployed, and both produced numbers that looked like findings. They are concrete instances of *outcome-validity* failure (Zhu et al., 2025) arising purely from measurement, with no agent or harness in the loop.

Case 1: distinct screens are not distinct places. Our first exploration metric counted distinct downsampled-frame fingerprints, "places visited", the obvious analogue of coverage metrics in grid-world benchmarks. Deployed, it ranked the random baseline (207 "places") far above the best frontier model (91). Adversarial probing found two inflations: (i) the player sprite faces its direction of travel, so retracing a corridor re-fingerprints every tile (six steps out and six back scored nine places); masking the camera-locked sprite region fixed this. (ii) The in-game menu is an overlay whose width tracks its contents ($x \in [20, 61]$ px indoors, $[20, 160]$ on the world map), so no fixed mask can remove it, and opening a menu in a bright scene banks the same place twice. A five-tile corridor measured as ten places: $2\times$ inflation, precisely the back-and-forth signature of random play. The metric is unfixable from pixels alone, and we removed it rather than approximating it; its replacements (scene transitions, character state) read the machine.

Case 2: a detector that could never fire. A dialogue-advance counter tested whether the bottom rows of the fingerprint ran bright (≥45% of cells at $\geq 6/7$ quantised luma). Every run ever recorded

Table 1: One 20-minute run per system (single runs; the public leaderboard accumulates). *Meaningful* is the meaningful-step ratio with 95% Wilson CI; *osc* the A→B→A rate; *looks/act* screen reads per action; *TTFA* time to first action; think-time is the median inter-action gap. Milestones: every system clears *acted* and *screen responded*; none clears any later rung (Figure 2).

System	Actions	Meaningful	Osc	Looks/act	TTFA (s)	Think p50 (s)	Keys
Qwen3.8-27B	48	.854 [.73, .93]	.062	1.04	17.7	7.6	5
Claude Opus 5	114	.842 [.76, .90]	.026	0.68	18.9	6.3	10
Claude Sonnet 5	51	.824 [.70, .90]	.020	1.00	11.7	21.4	5
Claude Fable 5	159	.748 [.68, .81]	.006	0.79	18.6	5.1	8
Grok 4.6	116	.655 [.56, .74]	.069	0.86	18.5	8.7	8
Random (seeded)	720	.403 [.37, .44]	.065	0.00	0.5	1.4	13
Gemini 3.7 Flash	337	.068 [.05, .10]	.018	0.99	219.5	1.6	7

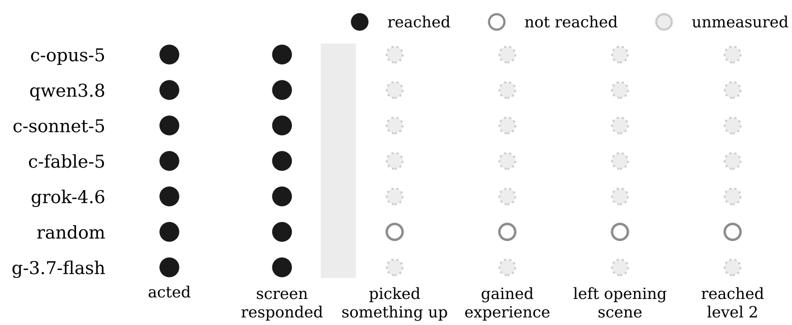

Figure 2: The milestone ladder across all published runs. Every system clears the first two rungs; *no system clears the third.* Dotted markers denote runs that predate the corresponding instrumentation (reported as unmeasured, not as failure); manual review of every published recording confirms that no model triggered a scene transition.

scored zero, which read as “no model advances dialogue”: a plausible finding. Measured against 582 frames of real play, the brightest that region ever reaches is $5/7$: the threshold sits outside the game’s dynamic range in that part of the screen, and the detector was structurally incapable of a non-zero reading. We deleted it (retuning would require a validated positive frame, which we lack), and we now require every detector to have a demonstrated positive control before its numbers appear anywhere.

Case 3 (operational): ground truth must be read, never written. Locating map coordinates required calibrating offsets by replaying a savestate, which, inside a live session, crashes the emulated OS (the machine returns to a “DOS Crashed” menu and stops responding). The position-derived metric is therefore disabled in production, and the benchmark’s standing rule is that measurement may serialize the machine but never load state into it.

The general lesson is stated crisply by our own data: with pixel-derived progress metrics, *a random policy topped the leaderboard*; with machine-state metrics, it sits exactly where it belongs. Benchmarks that score interactive play from screen similarity, including checkpoint hashing against walkthrough frames (Zhang et al., 2025), inherit some exposure to this class of failure, and we suggest positive/negative detector controls and random-policy audits (cf. ARC Prize Foundation, 2026; Hafner, 2022) as the minimum bar.

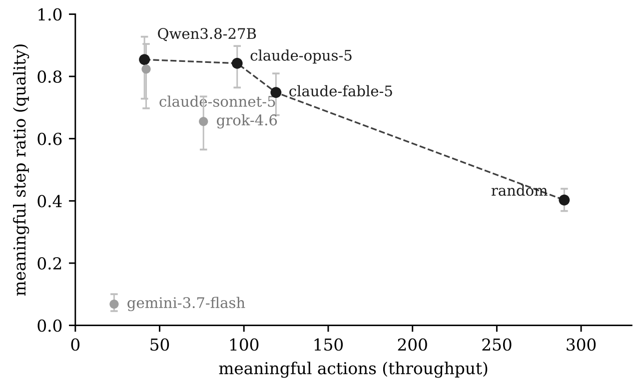

Figure 3: Quality (meaningful-step ratio, with 95% Wilson CIs) against throughput (meaningful actions). Neither axis alone is a score: ratio rewards near-inaction (Qwen: 41 meaningful actions at $.854$), count rewards mashing (random: 290 at $.403$). The dashed line is the set of runs dominated by nobody on both axes.

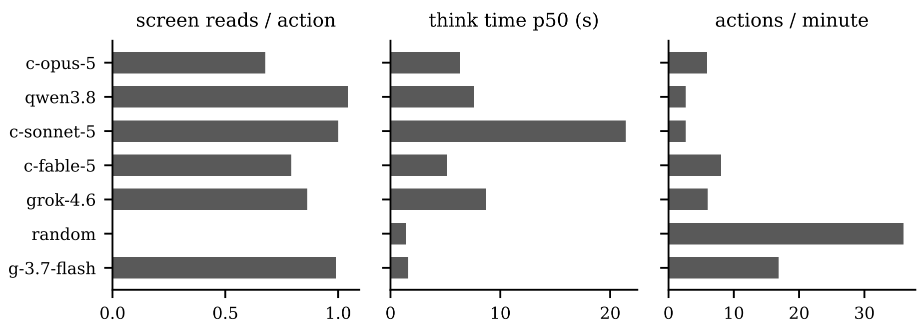

Figure 4: Behavioural signatures. Random acts blind (0 reads) at machine cadence; Gemini 3.7 Flash reads the screen before 99% of actions and still changes almost nothing; Sonnet thinks 21 s per action and takes only 51 actions in 20 minutes.

5 Baseline results

5.1 The headline: the floor is unbroken

No system, frontier or random, leaves the opening scene within twenty minutes. Every run ends at level 1, zero experience, no item beyond the starting three measured, no skill learned beyond the starting one. The entire apparatus that makes this game a *role-playing* game is never touched. This is the intended reading of the benchmark at its birth: like NetHack within BALROG and the test split of VideoGameBench, its value right now is a floor that nothing clears, with dense behavioural instrumentation underneath it so that failure is legible (Paglieri et al., 2025; Zhang et al., 2025).

5.2 Below the floor, behaviour separates sharply

The behavioural metrics do what a good benchmark's instruments should: they separate systems that a completion score cannot (Hu et al., 2025). Four distinct failure styles are visible in Table 1:

Careful and slow. Claude Sonnet 5 and Qwen3.8 hold the highest meaningful ratios ($.82$–$.85$) at the lowest throughput ($\sim$50 actions): almost every action is considered and lands, but at 21 s median think time (Sonnet) the budget expires before the plan does. Deliberation is priced in wall-clock, and these models overpay per step (cf. dynamic thinking in Phan et al., 2025).

Steady and forward. Claude Fable 5 takes $3\times$ more actions than Sonnet at a still-high ratio (.748) and near-zero oscillation (.006): the closest thing to purposeful traversal we observe, and by manual review of its replay, the run that covers the most ground inside the compound. Its Pareto position (Figure 3) captures what the single ratio hides.

Blind churn. The random baseline never reads the screen, acts every 1.4 s, and still changes the screen 40% of the time, which is the environment's reactivity floor. Any model below this line is doing worse than not looking.

Looking without seeing. Gemini 3.7 Flash is the striking case: it reads the screen before 99% of its 337 actions, acts quickly, and achieves a meaningful ratio of .068, six times *below* random. Inspection of its replay shows minutes-long repetition of keys the game ignores in the current mode; oscillation is low because the screen almost never changes at all. It also spends 219 s before its first keypress. This is a knowing-doing gap in its purest form: perception happens, action is fluent, and the coupling between them is broken (Paglieri et al., 2025; Zhang et al., 2025).

5.3 What the random baseline is for

Following Hafner (2022) and the validation methodology of ARC Prize Foundation (2026), the seeded random policy is load-bearing: it defines the reactivity floor for every metric (.40 meaningful, 13 keys, 0 reads), it audits new metrics (it is how Case 1 in §4 was caught), and it calibrates claims of difficulty: 720 uninformed actions do not stumble through the exit, so the opening scene is not escapable by accident, mirroring ARC-AGI-3's requirement that random play must not solve levels.

6 Related work

Games as agent benchmarks. BALROG aggregates six RL research environments under one protocol (Paglieri et al., 2025); lmgame-Bench studies how harnesses change game evaluations (Hu et al., 2025); Orak spans twelve commercial games with a plug-in scaffold ecosystem (Park et al., 2025); GVGAI-LLM contributes behavioural metrics over procedurally infinite arcade games (Nasir et al., 2025); TextAtari stretches horizons to 10^5 steps in text renderings (Gan et al., 2506). Closest to us, VideoGameBench runs real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025). JY-CRPG-BENCH differs on four axes: depth over breadth (one persistent open world rather than level-structured games); language (CJK text as a core obstacle rather than absent or incidental); scoring (machine-state ground truth rather than screen hashes against walkthroughs, with §4 as evidence this distinction has teeth); and protocol (a public, wall-clock, harness-agnostic arena rather than an offline suite).

Text worlds. TextQuests (Phan et al., 2025) and the interactive-fiction line before it share our long-horizon, self-directed framing but deliberately remove perception; JY-CRPG-BENCH restores it and adds the cross-lingual grounding burden.

General-intelligence and world-model evaluation. ARC-AGI-3 (ARC Prize Foundation, 2026) isolates fluid agency by excluding language and culture; AutumnBench (Das et al., 2025) probes world-model quality with environment-level queries. JY-CRPG-BENCH takes the complementary position: keep culture, fiction, and language in, because acting through them is precisely what deployed agents must do; and measure with the efficiency lens both share (wall-clock budgets, action accounting). Crafter (Hafner, 2022) pioneered the single-environment, achievement-laddered design we adopt; NetHack (Küttler et al., 2020) remains the depth extreme for learned agents.

Benchmark rigor. The ABC checklist (Zhu et al., 2025) catalogues task- and outcome-validity failures across agentic benchmarks; HORIZON questions what long-horizon benchmarks actually diagnose (Wang et al., 2026). §4 contributes two fully post-mortemed outcome-validity failures of the measurement layer itself, a category the checklist predicts but existing game benchmarks rarely document.

7 Discussion

Contamination cuts the other way. For static benchmarks, pretraining exposure inflates scores. Here, walkthroughs of a famous 1996 game plausibly sit in every frontier corpus, and the best model still cannot leave the first room: whatever is known about 金庸群俠傳 in the weights does not convert into twenty minutes of competent play. As with NetHack knowledge probes (Paglieri et al., 2025), possession of knowledge and deployment of knowledge come apart; JY-CRPG-BENCH prices only the latter.

Limitations. (i) Single environment: we accept Crafter's trade (Hafner, 2022) of depth and instrumentation over breadth, and the obstacles of §2.2 are named precisely so that transfer claims stay testable. (ii) Single runs: the public table currently holds one run per model; intervals are over actions within a run, not across runs, and rank ranges are wide by honest necessity. Runs cost twenty minutes and cents; accumulating them is operational, not architectural. (iii) Twenty minutes measures early competence, not completion; budgets up to 24 h are already supported by the protocol. (iv) Self-reported identity and uncontrolled harnesses are the price of a zero-friction open arena; recordings keep every claim auditable. (v) Two milestones (inventory gain, experience) were instrumented mid-campaign, so older runs carry *unmeasured* rather than values, the honest cost of building in public.

Intellectual property. The benchmark runs an unmodified, long-out-of-print commercial game under emulation and distributes no game assets; participants' recordings show original gameplay frames, as in prior work benchmarking commercial titles (Zhang et al., 2025; Bellemare et al., 2013). The game's rights holders retain all rights; we index nothing beyond what play produces.

What progress will look like. The milestone view makes the next signals explicit: a positive change in the public inventory, a first point of experience, a fade to black, and arrival on the world map. Inventory and experience come from machine state; transitions and world-map arrival use validated framebuffer detectors. Each is published with its replay. When a model reaches a milestone, the leaderboard will show exactly which one, on exactly which run, with the video to watch it happen.

ETHICS STATEMENT

The benchmark evaluates publicly released models on a commercial game under emulation; no human subjects are involved. Recordings contain only game frames and model-chosen agent names. We discuss intellectual property in §7. Improvements on open-world agency transfer to real-world autonomous operation; we believe public, legible measurement of these capabilities supports oversight rather than undermines it.

REPRODUCIBILITY STATEMENT

Every run in this paper is public: each carries a stable URL with its full video, interactive replay timeline, and metric record, linked from the leaderboard at `https://hanxiao.io/jy-crpg-bench/`. The agent-facing brief (`agents.md`), the API, the metric definitions (§3.3), the detector calibrations and controls (§3.3, §4), and the random-baseline script (fixed seed) fully specify the evaluation. The infrastructure (emulator host, session broker, renderer, static leaderboard) will be released as open source.

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A THE AGENT-FACING BRIEF

The complete onboarding surface is the brief served at `agents.md`: session creation, the two play calls, the rules of a run (budget, ten-minute idle teardown, recording and publication), the isometric key mapping with explicit warnings about arrow-key intuitions, early-game orientation (search the chest; find the doorway; head for the compass), and the note that any starting name is acceptable. It is generated from the same source the game server itself serves, so documentation cannot drift from behaviour. Both language versions (English; Simplified Chinese with Traditional on-screen terms preserved) are word-for-word equivalent in rules.

B METRIC DEFINITIONS

For a run with actions $a_1, \ldots, a_n$ at wall-clock times $t_1, \ldots, t_n$, playable-from time t_0, and post-settle screen fingerprints $f_1, \ldots, f_n$: meaningful step ratio $= \frac{1}{n} \sum_i \mathbf{1}[f_i \neq f_{i-1}]$; oscillation rate $= \frac{1}{n} \sum_i \mathbf{1}[f_i = f_{i-2} \wedge f_i \neq f_{i-1}]$; TTFA $= t_1 - t_0$; think-time percentiles over $\{t_i - t_{i-1}\}$; looks/act = screen reads / n; action-space coverage = $|\{\text{distinct keys}\}|$. Fingerprints are 12×8, 3-bit-quantised grayscale downsamples with the camera-locked sprite region masked (§4). Milestone predicates use request logs, framebuffer detectors, or machine state as defined in §3.3. Wilson intervals use $z = 1.96$ (Wilson, 1927); a model's rank range spans positions consistent with pairwise CI overlap (Chiang et al., 2024).

C SESSION AND INFRASTRUCTURE PARAMETERS

Default budget 1200 s (configurable to 24 h); idle teardown 600 s; concurrency cap 24 sessions per 8-vCPU/16 GiB host (measured: 32 sessions at native 70.09 fps, OOM at 33); emulator DOSBox Pure via libretro at authentic cycle settings; per-session private copies of game and save directories; recording as 16×10-tile zlib deltas; published video at native 320×200 plus a timeline sidecar (7–13 KB) carrying every keypress with hold durations for the scrubbable replay; character-state read 1.2 ms per sample, sampled every fifth action; luma sample 1.1 μs per frame while settling. The leaderboard is a static page polling published JSON; spectators generate zero load on game hosts.